

Paley Graphs as Exact Ihara Nonbacktracking Dynamics

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3 September 2026

Abstract

For every odd prime power $q \equiv 1 \pmod{4}$, we give one closed algebraic description of the Paley graph and its directed nonbacktracking dynamics. The graph is strongly regular with parameters $(q, (q-1)/2, (q-5)/4, (q-1)/4)$; additive characters give its complete three-point adjacency spectrum. A self-contained incidence-matrix elimination then produces the full Bass factorization, every Hashimoto eigenvalue, every power trace, and every oriented primitive-cycle count. The small face $q = 5$ is the five-cycle and has exactly two oriented prime circuits of length five. Exact computations reconstruct 13 fields, 25,901 adjacency cells, and 369,848 legal edge transitions. These are source-graph facts: the finite-field label is not a rational-prime orbit law, and the Ihara product is not a target Euler factor or an RH assertion.

Revision certificate. Bass spectrum and primitive-cycle closure.

1 Model and theorem

Let q be an odd prime power with $q \equiv 1 \pmod{4}$. Extend the quadratic character χ of $\mathbb{F}_q$ by $\chi(0) = 0$. The Paley graph $P(q)$ joins distinct x, y precisely when $\chi(x - y) = 1$. Put

$$k = \frac{q-1}{2}, \quad m = \frac{qk}{2}, \quad r = \frac{-1 + \sqrt{q}}{2}, \quad s = \frac{-1 - \sqrt{q}}{2}.$$

Its directed-edge matrix is

$$B_{(x,y),(y,z)} = 1 \iff y \sim z \text{ and } z \neq x.$$

Prime cycles are oriented and identified under cyclic shift, but not under reversal.

Theorem 1. *The graph $P(q)$ is connected and strongly regular with parameters*

$$\left(q, k, \frac{q-5}{4}, \frac{q-1}{4} \right),$$

and adjacency spectrum k^1, r^k, s^k . Its nontrivial spectrum obeys the source Ramanujan bound $\max(|r|, |s|) \leq 2\sqrt{k-1}$. Moreover

$$\begin{aligned} \det(I - uB) &= (1 - u^2)^{m-q} (1 - ku + (k-1)u^2) \\ &\quad \times (1 - ru + (k-1)u^2)^k (1 - su + (k-1)u^2)^k. \end{aligned} \tag{1}$$

Thus every eigenvalue of B is a root of $z^2 - \lambda z + (k-1)$ for $\lambda \in \{k, r, s\}$, with the adjacency multiplicity, together with $+1$ and -1 each $m - q$ additional times. If $N_n = \text{Tr}(B^n)$ and π_n counts oriented prime cycles of length n , then

$$N_n = \sum_{d|n} d \pi_d, \quad \pi_n = \frac{1}{n} \sum_{d|n} \mu(d) N_{n/d}, \quad Z_{P(q)}(u) = \det(I - uB)^{-1}.$$

For $q = 5$, the excess $m - q$ is zero and $\pi_5 = 2$.

2 Character proof

Because $\chi(-1) = 1$, adjacency is symmetric and every vertex has the k nonzero squares as increments. Let $J(t) = (1 + \chi(t) - \delta_0(t))/2$. For $a \neq 0$, expansion of the two indicators and the elementary identity $\sum_t \chi(t)\chi(t-a) = -1$ give

$$\sum_{t \in \mathbb{F}_q} J(t)J(t-a) = \frac{q-3-2\chi(a)}{4}.$$

This is $(q-5)/4$ on an edge and $(q-1)/4$ off an edge.

Fix a nontrivial additive character ψ of $\mathbb{F}_p$ and write $\psi_b(x) = \psi(\text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(bx))$. Convolution by the nonzero squares diagonalizes on these characters. The trivial eigenvalue is k , while for $b \neq 0$

$$\lambda_b = \frac{-1 + \chi(b)G}{2}, \quad G = \sum_t \chi(t)\psi(\text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(t)), \quad G^2 = \chi(-1)q = q.$$

The sign of G can exchange the labels r, s but not their multiset. Half of the nonzero b have each sign, proving k^1, r^k, s^k . Since the degree eigenvalue of a regular undirected graph has multiplicity equal to its number of connected components, $P(q)$ is connected. Finally,

$$\max(|r|, |s|) = \frac{\sqrt{q} + 1}{2} \leq \sqrt{2(q-3)} = 2\sqrt{k-1}.$$

After squaring this is $2\sqrt{q} \leq 7q - 25$; the difference is increasing for $q \geq 5$ and equals $10 - 2\sqrt{5} > 0$ at five. This is only the Ramanujan bound for the source graph.

3 Bass elimination and the entire edge spectrum

Let S, T be tail and head incidence matrices from vertices to directed edges, and let R reverse a directed edge. With our indexing,

$$B = T^*S - R, \quad R^2 = I, \quad SR = T, \quad ST^* = A, \quad SS^* = kI.$$

Factor $I - uB = (I + uR)[I - u(I + uR)^{-1}T^*S]$. Since $(I + uR)^{-1} = (I - uR)/(1 - u^2)$, the identity $\det(I - XY) = \det(I - YX)$ reduces the second determinant to vertex space. Multiplication by $1 - u^2$ and the displayed incidence identities give $I - uA + (k-1)u^2I$. Finally, R is a direct sum of m swaps, so

$$\det(I - uB) = (1 - u^2)^{m-q} \det(I - uA + (k-1)u^2I).$$

Substituting the adjacency spectrum proves (1). Its degree is $2(m-q) + 2q = 2m$, hence no algebraic eigenvalue or multiplicity is omitted.

The entry $(B^n)_{e,e}$ counts based, closed, tailless nonbacktracking walks. A prime cycle of length $d \mid n$ contributes its d possible base edges to the n/d repetition. This proves $N_n = \sum_{d \mid n} d\pi_d$; Möbius inversion gives π_n . The formal trace-log identity

$$-\log \det(I - uB) = \sum_{n \geq 1} \frac{\text{Tr}(B^n)}{n} u^n$$

gives the source Ihara zeta and its prime-cycle product. When $q = 5$, the nonzero squares are $\{1, 4\}$, so the graph is C_5 . A nonbacktracking walk must preserve direction; its two orientations give $\pi_5 = 2$.

Verified sources

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